Introduction to Computer Science

18th November 2024

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

SECTION A - Multiple Choice Question
Circle ONE (1) correct answer from A, B, C, or D for each question.
Each question is worth 1 mark.

Question 1
What is an embedded system?

A A computer system not linked to another computer system

B A device that contains a computer, such as a car or washing machine

C A computer system that undertakes most frequent tasks required by general users

D Two or more computer systems linked together

1 mark

Mark Scheme: B: A device that contains a computer, such as a car or washing machine

Question 2
What is the purpose of a fan on a CPU?

A To speed up the CPU's processing power

B To increase the computer's overall storage capacity

C To reduce the noise made by the CPU

D To prevent the CPU from overheating

1 mark

Mark Scheme: D: To prevent the CPU from overheating

Question 3
What does a computer's clock regulate and how is its speed measured?

A It regulates memory storage and is measured in bits per second

B It controls CPU cooling and is measured in watts

C It regulates the timing and speed of all functions, measured in Megahertz (MHz) or Gigahertz (GHz)

D It manages data transmission between devices and is measured in gigabytes

1 mark

Mark Scheme: C: It regulates the timing and speed of all functions, measured in Megahertz (MHz) or Gigahertz (GHz)

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Marks

Question 4
What is Network Attached Storage (NAS)?

A A device directly connected to a computer to store data

B A network hard drive that can store, retrieve and share data over a network

C A type of cloud storage used only for businesses

D A storage system that cannot be accessed from home

1 mark

Mark Scheme: B: A network hard drive that can store, retrieve and share data over a network

Question 5
What is software piracy?

A Legally purchasing software and sharing it with friends

B Unauthorised use, reproduction or distribution of software for profit

C Using free software provided by the developer

D Downloading software from a licensed vendor

1 mark

Mark Scheme: B: Unauthorised use, reproduction, or distribution of software for profit

Question 6
What is an operating system that allows more than one program to run simultaneously called?

A Multiprocessing operating system

B Multithreading operating system

C Real-time operating system

D Multitasking operating system

1 mark

Mark Scheme: D: Multitasking operating system

Question 7
What is the result of an XOR operation if both input bits are the same?

A 0

B 1

C The sum of the input bits

D The product of the input bits

1 mark

Mark Scheme: A: 0

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Question 8
How does a WAP function within a network?

A It directly connects devices to the internet.

B It acts as a firewall for all connected devices.

C It creates a wireless network and forwards traffic to the main wired network.

D It stores data from wireless devices

1 mark

***Mark Scheme: C**: It creates a wireless network and forwards traffic to the main wired network.*

Question 9
What is the difference between download and upload speeds in broadband internet?

A Download speed measures how fast you can receive data, and upload speed measures how fast you can send data.

B Download speed measures how fast you can send data, and upload speed measures how fast you receive data.

C Upload speeds are always faster than download speeds.

D Download speeds are measured in Mb/s. Upload speeds are measured in Kb/s

1 mark

Mark Scheme: A**: **Download speed measures how fast you can receive data, and upload speed measures how fast you can send data.

Question 10
How does a SQL injection attack compromise a website or application?

A By manipulating the query to gain unauthorised access to the database

B By crashing the server hosting the application

C By stealing data directly from the web server's file system

D By altering the website's appearance without database interaction

1 mark

Mark Scheme: A: By manipulating the query to gain unauthorised access to the database

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Marks

Question 11
What is ‘big data’?

A Data that requires to be saved on optical storage

B An increase in the volume of data stored and analysed by computer systems

C A type of cloud storage used for small data sets

D A method of processing data used in network protocols

1 mark

Mark Scheme: B: An increase in the volume of data stored and analysed by computer systems

Question 12
How does increasing a computer's RAM improve its performance?

A By speeding up the internet connection

B By improving the graphics rendering speed

C By enhancing the computer’s ability to run multiple tasks simultaneously

D By increasing the storage capacity for files

1 mark

Mark Scheme: C: By enhancing the computer’s ability to run multiple tasks simultaneously

Question 13
What is a HDMI cable primarily used for?

A Transmitting internet data between devices

B Connecting devices for data storage

C Transmitting digital audio and video between devices

D Powering electronic devices

1 mark

Mark Scheme: C: Transmitting digital audio and video between devices

Question 14
Which register holds the memory address of the next instruction to be fetched from primary memory?

A Memory Address Register (MAR)

B Program Counter (PC)

C Memory Data Register (MDR)

D Accumulator (ACC)

1 mark

Mark Scheme: B: Program Counter (PC)

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Marks

Question 15
What is the purpose of defragmentation software?

A It deletes old files from the hard drive to free up space.

B It compresses files to save storage space.

C It backs up files to prevent data loss.

D It rearranges file segments so they are stored contiguously, improving performance.

1 mark

Mark Scheme: D: It rearranges file segments so they are stored contiguously, improving performance.

Question 16
What is the decimal equivalent of the binary number 11100000?

A 128

B 192

C 224

D 256

1 mark

Mark Scheme: C: 224

Question 17
What is a key characteristic of bitmap (raster) images?

A They are made up of pixels arranged in horizontal rows

B They use mathematical equations to represent shapes.

C They cannot be edited unless converted to a PNG first

D They are unsuitable for photographs and image retouching.

1 mark

Mark Scheme: A**: **They are made up of pixels arranged in horizontal rows

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Marks

Question 18

Which of the following best describes bit depth in digital sound?

A The number of bits used per second, measured in kilobits per second (kbps).

B The number of sound samples taken per second.

C The number of bits available for each sample, with higher bit depth resulting in better sound quality.

D The number of audio channels in a recording.

1 mark

Mark Scheme: C: The number of bits available for each sample, with higher bit depth resulting in better sound quality.

Question 19

What is the value of output Q if the three inputs (R,S,T) in the logic statement below all have the initial value of (ONE) 1?

Q = NOT(R) AND (S OR T)

A 0

B 1

C 2

D -1

1 mark

Mark Scheme: A: 0

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Marks

Question 20
Which of the following is an example of cloud storage?

A Google Drive

B Local Hard Drive

C USB Flash Drive

D DVD

1 mark

Mark Scheme: A: Google Drive

Question 21
Which of the following correctly demonstrates the input, process and output in a supermarket checkout system?

A Input: The cashier counts the items manually.
Process: The computer calculates the total.
Output: The receipt is printed.

B Input: The computer finds the total.
Process: The cashier scans the items.
Output: The total is displayed on the screen.

C Input: Staff use a scanner to input the item.
Process: The computer finds the price from a database and adds it to a total.
Output: The total is shown on the screen.

D Input: The customer pays for the items.
Process: The cashier checks for discounts.
Output: The computer updates its inventory.

1 mark

Mark Scheme: C: Input: Staff use a scanner to input the item.
Process: The computer finds the price from a database and adds it to a total.
Output: The total is shown on the screen.

Question 22
How is a bus organised in a computer system?

A It only transmits addresses between components.

B It uses control lines for sending data and data lines for commands only.

C It has control lines for signalling and data lines for transmitting data and addresses.

D It does not include a process for sending or receiving data.

1 mark

Mark Scheme: C: It has control lines for signalling and data lines for transmitting data and addresses

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Marks

Question 23
Which type of wearable device is specifically designed to be worn on the wrist and offers features like fitness tracking and app notifications?

A VR Headset

B Hearables

C Smart Clothing

D Smartwatches

1 mark

Mark Scheme: D: Smartwatches

Question 24
What is a processor, also known as a central processing unit (CPU)?

A The main memory of the computer

B The main chip in a computer system located on a motherboard

C A device used for cooling the computer

D A storage device for saving data

1 mark

Mark Scheme: B: The main chip in a computer system located on a motherboard

Question 25
What is the effect of having more cores in a CPU?

A It doubles the number of instructions that can be processed.

B It allows the CPU to process more instructions, but not necessarily double.

C It reduces the processing power of the CPU.

D It decreases the number of fetch-decode-execute (FDE) cycles.

1 mark

Mark Scheme: B: It allows the CPU to process more instructions, but not necessarily double.

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Marks

Question 26
What is the smallest unit of measurement in computing?

A Nibble

B Kilobyte

C Byte

D Bit

1 mark

Mark Scheme: D: Bit

Question 27
Which of the following is an example of a second-generation programming language?

A Binary machine code

B Python

C JavaScript

D Assembly Language

1 mark

Mark Scheme: D: Assembly Language

Question 28
What is the purpose of Plug and Play in modern operating systems?

A It allows users to manually configure new hardware.

B It enables automatic detection and configuration of new hardware peripherals.

C It disables hardware detection to increase system speed.

D It automatically updates all software installed on the computer.

1 mark

Mark Scheme: B: It enables automatic detection and configuration of new hardware peripherals

Question 29
What is the process for representing a negative integer in two's complement notation?

A Write the number in binary, invert the bits and add 1

B Invert the number and subtract 1

C Write the number in binary and add 1 to the leftmost bit

D Subtract the binary number from 11111111

1 mark

Mark Scheme: A: Write the number in binary, invert the bits and add 1

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Marks

Question 30

Why is Unicode more suitable for representing characters from different languages compared to older character encoding systems?

A It uses 8 bits and supports 256 characters.

B It only supports English and a few European languages.

C It supports over 65,000 characters, making it suitable for multiple languages.

D It requires less memory than ASCII and includes emojis.

1 mark

Mark Scheme: C: It supports over 65,000 characters, making it suitable for multiple languages

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Marks

SECTION B
ANSWER ALL QUESTIONS

Question 31

a) A hospital vending machine provides hot and cold drinks.

Customers select their choice of drink from a menu on screen and provide payment through a chip and pin card reader.

When payment has been taken the machine dispenses the drink and displays a message on the screen thanking the customer.

i) Describe the vending machine system in terms of input, process and output. **3**

Mark Scheme:

Input:

- ***Customer selects a drink from the menu on the screen using the touch screen to input their selection.***
- ***Customer provides payment through the chip and pin card reader (input of payment details).***

Process:

- ***The vending machine processes the customer's selection by retrieving the drink option.***
- ***It processes the payment by validating and charging the customer's card through the chip and pin reader.***
- ***If payment is confirmed, the system prepares the drink.***

Output:

- ***The vending machine dispenses the chosen hot or cold drink.***
- ***A message is displayed on the screen thanking the customer for their purchase.***

1 mark for any bullet point per section, max 1 mark per section.

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Marks

ii) State the type of interface the vending machine uses to interact with customers (shown below). **1**



Mark Scheme:

- ***Graphical User Interface (GUI)***

1 mark

iii) State the name of TWO (2) INPUT devices used in the system. **2**

Mark Scheme:

- ***Touchscreen or Button Panel – used to select the drink from the menu.***
- ***Chip and Pin Card Reader – used to input payment information.***

1 mark for each bullet

iv) Last year the hospital employed a person to serve the drinks. **4**

Describe ONE (1) advantage and ONE (1) disadvantage of replacing the human worker with the drinks vending machine.

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Marks

Mark Scheme:

Advantages:

- ***Cost Reduction (1 mark): No need to pay wages, benefits or manage labour-related expenses – holiday pay (2nd mark)***
- ***24/7 Availability: The vending machine can operate continuously, providing drinks outside of typical working hours without needing breaks or rest.***
- ***Consistency and Speed: Machines provide consistent service and dispense drinks quickly, reducing wait times and human error.***
- ***Reliability: Machines won't get ill or be late for work***

Disadvantages:

- ***Limited Customisation (1 mark): A vending machine offers a set menu and cannot handle special requests or customised drink orders, unlike a human worker (2nd mark).***
- ***Technical Failures: Machines can malfunction or break down, requiring maintenance, which may cause delays in service until repaired.***
- ***No Personal Interaction: Patients and visitors who may prefer human interaction or need assistance may find the machine impersonal and less accommodating.***
- ***Initial Setup Costs: Purchasing, installing and maintaining a vending machine may involve significant initial costs and ongoing upkeep expenses.***
- ***Potential job loss***

1 mark for a simple statement in each section x 2, 1 mark for <u>description</u> in each section x 2 (See first bullet example in each)

Total 10 Marks

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Marks

Question 32

a) A police station records all suspect interviews as audio files with maximum file size 100MB.

i) Calculate the maximum number of audio files they could store on a 1TB external hard drive. You must show your working. **2**

Mark Scheme:

- ***Convert 1TB to MB:***
- ***1TB = 1,000 GB***
- ***1GB = 1,000 MB***
- ***Therefore, 1TB = 1,000,000 MB.***
- ***1,000,000 MB (storage) ÷ 100 MB (file) = 10,000 files.***

Thus, the police can store 10,000 audio files on a 1TB external hard drive

1 mark for 10'000. 1 mark for working out.

ii) Explain the term 'solid-state' memory. **2**

Mark Scheme:

- ***It is a type of data storage that uses non-volatile memory chips***
- ***It is solid as it does not have mechanical parts like spinning disks and read/write heads***
- ***This makes it faster and more durable***

1 mark per bullet to a max of 2

iii) State ONE (1) disadvantage of using solid state storage compared to magnetic tape. **1**

Mark Scheme:

- ***Higher Cost: Solid-state storage is generally much more expensive per gigabyte than magnetic tape, making it less cost-effective for storing large amounts of data.***
- ***Less Suitable for Long-term Archiving: Magnetic tape is specifically designed for long-term, large-scale data archiving, as it can store data reliably for decades, whereas solid-state storage may degrade over time without regular maintenance.***

1 mark for any bullet (no detail needed)

iv) The police have decided to try Cloud storage. **4**

Describe ONE (1) advantage **and** ONE (1) disadvantage of using Cloud storage.

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Marks

Mark Scheme:

Advantages:

- ***Scalability: Cloud storage allows the police to easily scale their storage capacity as needed without having to invest in additional physical hardware.***
- ***Remote Accessibility: Cloud storage enables police staff to access files and data from anywhere with an internet connection, which can improve collaboration and productivity***
- ***Cost Efficiency: Cloud storage typically operates on a pay-as-you-go model, reducing the need for large upfront investments in hardware and maintenance costs.***

Disadvantages:

- ***Dependency on Internet Connection: Cloud storage requires a stable internet connection to access or upload files. If the police experiences network outages or poor connectivity, access to important data can be disrupted***
- ***Security and Privacy Concerns: Storing sensitive police data in the cloud may raise concerns about data breaches and unauthorised access***
- ***Limited Control: By using a third-party cloud service, the police rely on the provider for the management, security and availability of their data. This means less control over their data infrastructure***

1 mark for a simple statement in each section x 2,
1 mark for <u>description</u> in each section x 2

v) The audio of each interview is converted into a lossy audio file. **1**

State ONE (1) possible file format for the recordings.
Mark Scheme:

- ***MP3 (.mp3)***
- ***AAC (.aac)***
- ***WMA (.wma)***
- ***OGG (.ogg)***
- ***M4A (.m4a)***

1 mark for any bullet

Total 10 Marks

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Marks

Question 33

a) Explain what is meant by the term ‘software’? **1**

Mark Scheme:

- ***A collection of programs, data and instructions that tell a computer or electronic device how to perform specific tasks***

1 mark

b) For each example of software given below state whether it is application software or utility software.

i) CAD **1**

Mark Scheme:

- ***Application Software***

1 mark

ii) E-Mail **1**

Mark Scheme:

- ***Application Software***

1 mark

iii) Web Browser **1**

Mark Scheme:

- ***Application Software***

1 mark

iv) Anti-virus **1**

Mark Scheme:

- ***Utility Software***

1 mark

c) Describe how an Operating System supports the following aspects of computer usage:

i) Interrupt Handling **2**

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Marks

Mark Scheme:

- ***The OS uses interrupts to handle unexpected or high-priority events that require immediate attention, such as input from a user or hardware failure.***
- ***When an interrupt occurs, the OS pauses the current process, saves its state and transfers control to handle the event.***
- ***After the interrupt is handled, the OS restores the previous process and continues its execution, ensuring smooth and responsive performance.***

1 mark per bullet to a max of 2

ii) Managing Memory **2**

Mark Scheme:

- ***The OS is responsible for allocating memory for different programs and processes.***
- ***It uses paging to ensure that each process gets the memory it needs while preventing conflicts between programs.***
- ***If the system runs out of physical memory, the OS can use virtual memory, where part of the hard drive is used as an extension of RAM, allowing for more programs to run simultaneously.***

1 mark per bullet to a max of 2

d) State ONE (1) feature of an Operating System that uses a WIMP Interface. **1**

Mark Scheme:

- ***Uses Windows***
- ***Contains Clickable Icons***
- ***Uses Menus***
- ***Will be controlled by a Pointer***

1 mark for any bullet

Total 10 Marks

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Marks

Question 34

a) Complete the table by converting the 8-bit binary number into decimal and the decimal into 8-bit binary. **3**

8-bit Binary	Decimal
11110000	
	106
00001111	

Mark Scheme:

8-bit Binary	*Decimal*
11110000	***240 (1 mark)***
0110 1010 (1 mark)	***106***
00001111	***15 (1 mark)***

b) Complete the table by writing the answer for each statement. **3**

Statement		Answer
(i)	The binary code that represents the hexadecimal value ‘B’	
(ii)	The smallest hexadecimal value that can be represented using 5-bits	
(iii)	The maximum number of different colours that can be represented in a PNG with a colour depth of 7 bits	

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Marks

Mark Scheme:

Statement	***Answer***
(i) The binary code that represents the hexadecimal value 'B'	***B=11 so in binary 11= 1011 Accept 00001011 (1 mark)***
(ii) The lowest hexadecimal value that can be represented using 5-bits	***Binary 00000 = Hex 0 0 (1 mark)*** ***There must only be 2 zeros as answer is in hex***
(iii) The maximum number of different colours that can be represented in a PNG with a colour depth of 7 bits	***128 (1 mark)***

c) Perform a TWO (2) place binary shift to the left on the value 00011010 and describe the mathematical effect of this operation. **2**

Mark Scheme:

- ***0110 1000***
- ***Multiplication by 4***

1 mark per bullet

d) Explain why binary addition may lead to an overflow error. **2**

Provide an example to support your answer.

Mark Scheme:

- ***An addition may need more bits than the CPU was expecting***
- ***Example should show two large 8-bit numbers being added together where a 9th bit would be required to store the addition (e.g 1111 1111 + 1111 1111)***

1 mark per bullet

Total 10 Marks

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Marks

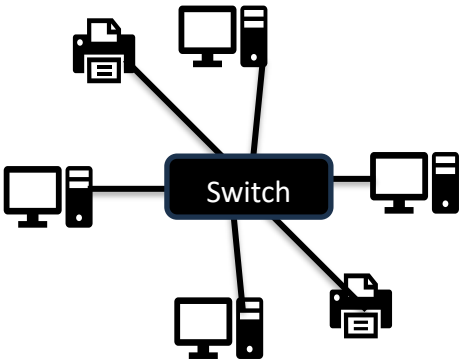
Question 35

a) A new office has installed a network which includes FOUR (4) computers and a switch. There are TWO (2) printers that can be accessed by all the computers.

The computers are arranged in a star topology.

i) Draw a diagram to show how the computers, printers and switch are connected in a star topology. **3**

Mark Scheme:



- ***Contains a single switch centrally***
- ***4 PC's have a single connection to the switch***
- ***2 printers have a single connection to the switch (with no connection directly to a PC)***

1 mark per bullet

ii) Describe the role of the switch in a star network. **2**

Mark Scheme:

- ***Receives data from a connected device***
- ***Forwards data to the specific device it is intended***
- ***Examines the destination MAC address in the data packet and sends it only to the appropriate device***
- ***Prevents data collisions***

1 mark per bullet to a max of 2

iii) State ONE (1) advantage of using a star topology over a ring topology. **1**

Mark Scheme:

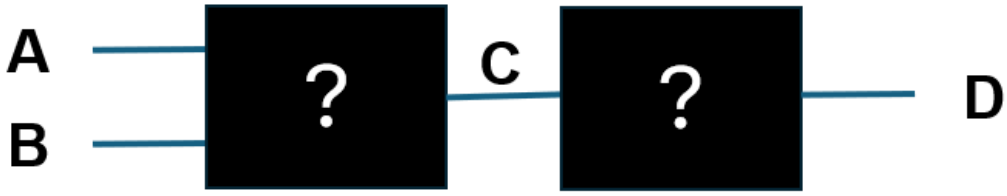
- ***Easier Troubleshooting and Maintenance***
- ***Better Scalability***
- ***Improved Performance***

1 mark for any bullet

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b) A logic diagram is given below without the logic gates shown.

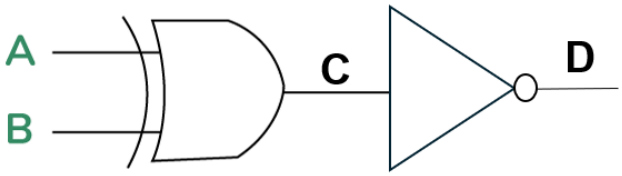
The truth table for the diagram is also provided.



Input A	Input B	C	D
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

i) Redraw the logic diagram using the correct symbols in place of the two unknown boxes. **3**

Mark Scheme:



- ***Box 1 shows an XOR gate symbol***
- ***Box 2 shows a NOT gate symbol (with circle)***
- ***All inputs are clearly labelled***

1 mark per bullet point.

ii) Describe in words the input conditions that would turn on a system that uses this logic sequence. **1**

Mark Scheme:

- ***The system is TRUE/ON if the two inputs are the same (1 mark)***

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Marks
Total 10 Marks

Question 36

a) State TWO (2) ethical issues in Computing. **2**

Mark Scheme:

- ***Privacy***
- ***Surveillance***
- ***Digital Divide***
- ***Intellectual Property***
- ***Cybersecurity***
- ***Automation and Job Displacement***
- ***AI and Machine Learning Bias***
- ***Digital Addiction and Well-being***

1 mark per suitable answer (not restricted to bullets)

b) Describe TWO (2) ways the growth of computer usage has had a **positive** impact on the environment. **4**

Mark Scheme:

- ***Reduced Paper Usage (1 mark): The digitalisation of documents, communications and record-keeping has significantly reduced the need for physical paper. E-books, online reports and digital communication (like email) reduce deforestation and lower the environmental impact of paper production (1 mark).***
- ***Remote Work and Telecommuting (1 mark): The ability to work from home, enabled by computers and the internet, reduces the need for commuting, leading to lower carbon emissions from transportation. This also reduces energy consumption in office buildings (1 mark).***
- ***Smart Technologies for Energy Efficiency (1 mark): Computers and digital technologies have enabled the development of smart systems for homes, businesses, and cities that optimise energy usage. For example, smart thermostats, lighting systems, and appliances help conserve energy by adjusting based on real-time data (1 mark).***
- ***Environmental Monitoring and Research (1 mark): Computers play a critical role in environmental science by helping researchers collect, analyse, and model data related to climate change, pollution, and conservation. This data helps governments and organisations take more effective action to protect the environment(1 mark).***

2 marks per bullet point, up to 4 marks

c) Employers are often concerned with the health of their employees if they are sat using a computer for long periods of time.

i) Describe ONE (1) health problem associated with working on computers for a long period of time. **2**

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Marks

Mark Scheme:

- ***Eye Strain (Computer Vision Syndrome): Prolonged screen time can cause eye strain, leading to symptoms like dry eyes, blurred vision, headaches and discomfort.***
- ***Repetitive Strain Injury (RSI): This can occur from performing repetitive tasks, such as typing or using a mouse, over extended periods. It can lead to pain, numbness or tingling in the hands, wrists, arms or shoulders, potentially leading to conditions like carpal tunnel syndrome.***
- ***Back and Neck Pain: Poor posture while sitting at a computer for long hours can lead to back, neck and shoulder pain.***
- ***Sedentary Lifestyle-Related Issues: Sitting for long periods without physical activity can contribute to various health problems, including obesity, cardiovascular issues and a higher risk of conditions like diabetes.***

1 mark for naming health issue, 1 mark for description

ii) State TWO (2) ways in which users can prevent health problems when using a computer. **2**

Mark Scheme:

- ***Regular breaks***
- ***Good posture***
- ***Adjust Monitor/Chair Height***
- ***Ergonomic setups***
- ***Physical movement***

1 mark for any bullet

Total 10 Marks

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Marks

Question 37

a) Describe how the World Wide Web works. **8**

You must include the following terminology in your answer:

- URL
- DNS
- IP address
- Web Server
- HTML

Mark Scheme:

- ***Main points to be covered: When a user enters a URL, the DNS translates the domain name into an IP address, which allows the browser to locate the server hosting the requested webpage. The server responds by sending the page's HTML content, which the browser renders for the user to view. This process happens quickly, allowing seamless browsing across the World Wide Web.***

- ***URL (Uniform Resource Locator): Every web page on the World Wide Web has a unique address known as a URL. When a user wants to access a webpage, they type the URL into a web browser, which specifies the location of the resource on the internet.***
- ***DNS (Domain Name System): After the URL is entered, the web browser needs to translate the human-readable domain name into an IP address. This translation is done by the DNS, mapping domain names to their corresponding IP addresses.***
- ***IP Address: The IP address uniquely identifies the server where the website is hosted. The browser sends a request to this IP address to retrieve the webpage.***
- ***HTML (Hypertext Markup Language): The web server hosting the website responds by sending the requested web page, which is written in HTML. The browser interprets the HTML code to display text, images, links, and other elements on the screen, allowing the user to view and interact with the webpage.***

Guidelines for Grading

- ***Comprehensive Discussion (7-8 marks): The candidate covers all areas with depth and balance, using full correct terminology and sequence. All 5 bullet points are used accurately.***
- ***Moderate Discussion (4-6 marks): The candidate addresses most areas but may lack depth in one or two, they use some correct terminology however answers shows minor misconceptions or emissions. All 5 bullet points are used but may not be accurate.***
- ***Limited Discussion (1-3 marks): The response mentions limited details with often incorrect terminology. There is little to no inclusion of the 5 bullets***

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Marks

b) Explain the term ‘phishing’ in relation to network security. **2**

Mark Scheme:

- ***A type of cyberattack where attackers attempt to trick individuals into revealing sensitive information, such as passwords, credit card details or other personal data.***
- ***This is done by impersonating a legitimate organisation or trusted entity through deceptive emails, websites or messages.***

1 mark per bullet

Total 10 Marks

End of paper

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Learning Outcomes matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
Q1	LO1	Yes
Q2	LO1	Yes
Q3	LO1	Yes
Q4	LO2	Yes
Q5	LO3	Yes
Q6	LO3	Yes
Q7	LO5	Yes
Q8	LO6	Yes
Q9	LO6	Yes
Q10	LO6	Yes
Q11	LO1	Yes
Q12	LO1	Yes
Q13	LO4	Yes
Q14	LO2	Yes
Q15	LO3	Yes
Q16	LO4	Yes
Q17	LO4	Yes
Q18	LO4	Yes
Q19	LO5	Yes
Q20	LO2	Yes
Q21	LO1	Yes
Q22	LO1	Yes
Q23	LO1	Yes
Q24	LO1	Yes
Q25	LO1	Yes
Q26	LO4	Yes
Q27	LO2	Yes
Q28	LO2	Yes
Q29	LO4	Yes
Q30	LO4	Yes
31	LO1 L07	Yes
32	LO2	Yes
33	LO3	Yes
34	LO4	Yes
35	LO5 LO6	Yes
36	LO7	Yes
37	LO6	Yes

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Marks

Grade descriptors

Learning Outcome	Pass	Merit	Distinction
Understand fundamental concepts relating to computer systems	Demonstrate adequate level of understanding	Demonstrate robust level of understanding	Demonstrate highly comprehensive level of understanding
Understand the characteristics of secondary storage	Demonstrate adequate level of understanding	Demonstrate robust level of understanding	Demonstrate highly comprehensive level of understanding
Understand application and system software	Demonstrate adequate level of understanding	Demonstrate robust level of understanding	Demonstrate highly comprehensive level of understanding
Understand how computers store data	Demonstrate adequate level of understanding	Demonstrate robust level of understanding	Demonstrate highly comprehensive level of understanding
Understand logic gates	Demonstrate adequate level of understanding	Demonstrate robust level of understanding	Demonstrate highly comprehensive level of understanding
Understand the fundamental concepts of computer networks and threats to network security	Demonstrate adequate level of understanding	Demonstrate robust level of understanding	Demonstrate highly comprehensive level of understanding
Understand Ethical, Environmental and Cultural issues in Computing	Demonstrate adequate level of understanding	Demonstrate robust level of understanding	Demonstrate highly comprehensive level of understanding

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